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RESEARCH-ARTICLE

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Persona-driven approach to enhance security requirement elicitation

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Abstract

Security Requirements Engineering (SRE) is a fundamental process in the development of secure systems, ensuring that security needs are identified and addressed early in the software development lifecycle. However, existing literature highlights a gap in integrating human aspects of security within SRE, which can lead to complex security solutions that are prone to human errors, increasing the risk of security breaches. Over the years, the need to incorporate user-centric approaches into security design has become apparent. This paper presents a persona-based approach that incorporates human factors in the early stages of system development and thereby improves the quality of the security requirements elicitation process. By characterizing users and attackers based on their needs, motivations, and behaviors, the approach enhances the identification of security vulnerabilities arising from human interactions. The proposed methodology contributes to both research and practice by enabling security engineers to design security mechanisms that align with user behavior, ultimately improving user security engagement and reducing security risks.

CCS Concepts

• **Security and privacy → Software security engineering**; • **Human and societal aspects of security and privacy → Usability in security and privacy**.

Keywords

Requirements Elicitation, Personas, Security Requirements, Human-Centric Engineering, Usable Security

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1 Introduction

Organizations spend considerable resources on developing usable and secure systems [24], [6], [15]. Nevertheless, security vulnerability remains a serious issue. Studies on data breaches reveal that a significant number of the breaches can be attributed to human factors. In 2024 [40], for instance, 68% of data breaches and successful attacks involving the human element. The need to address human factors adds to the technical challenges faced by the software engineers during software development. Within the systems development lifecycle, security is treated as a separate functionality, and techniques and development standards are available that focus on the technical perspectives of security. Human factors, however, often receive insufficient consideration during the systems' security development [22].

Existing literature on developmental practices indicate that security is often considered at later stages in the development lifecycle, which leaves less time with in the project's timeline to consider aspects other than technical security [7], [30], [31]. Since software development is a complex process that takes a long time and includes multiple stages, the initial requirements elicitation and analysis plays a significant role in ensuring a favorable outcome [33]. Elicitation of security requirements during the initial phases of software development can help identify potential threats and vulnerabilities, including those related to the human factors, enabling the management of these concerns and thereby reducing available avenues for data breaches caused by human errors [35].

To address the challenge posed by human behavior aspects of system security, the paper proposes that the human factors of security be considered as early as the requirements elicitation phase of the development lifecycle. Previous studies have also drawn attention to the need for consideration of human related factors in security requirements elicitation, for example [30], [3], [2], to our knowledge, however, none of these works discuss how to gather the security requirements from a diverse group of prospective users. Personas as a fictional representations of diversity user groups can be used as a design tool not only with providing a description of user needs but also by representing security behavior and needs from both the user and attacker perspectives [5], [27], [28]. By implementing personas, security engineers can identify behavior patterns, human vulnerabilities and potential risks, and they can thus model threat scenarios effectively, enabling them to create preventive strategies [32]. A persona approach can help development teams understand

how to make easy-to-use software and avoid system vulnerabilities and human errors which lead to data breaches. The aim of this paper is to present a persona-based approach for security requirements elicitation in software development that incorporates human factors. The technique enables the identification of new security requirements and refinement of existing requirements by fostering better understanding of groups or individuals with different behaviors. To achieve this objective, the research question (RQ) was formulated as follows:

RQ: *How can a persona-approach enhance security requirements elicitation in software development?*

The rest of the paper is organized as follows: in Section 2 the paper describes the motivation for the research and reviews related work relevant to human aspects in security engineering, enabling a better understanding of personas and its potential benefits in requirement engineering. Section 3 presents the proposed persona-based approach focusing on enhancing security requirements in software development process. Section 4 outlines the possible implementation of the findings based on a practical example and highlights how this approach addresses human, security and usability aspects. Future work and limitations are described in Section 5.

2 Related work in security requirements engineering

This section reviews related work on human factors in requirements engineering, especially the importance of using human-centric approaches in requirements elicitation and the role of persona implementation in the development of secure systems. The section ends by describing the research gap and the motivation for the work.

2.1 Human factors in RE

As noted earlier, the focus during requirements engineering (RE) should include identification of human aspects [35]. For instance, RE is dependent on the human factors by involving different participants in the process such as developers, business analysts, stakeholders and security engineers [16]. Human aspects such as personality, emotions, culture and gender positively influence RE effectiveness and support more accurate requirements through better understanding of end-users [16], [4]. Aspects such aspects as geographical and cultural, attitude and behavioral differences impact on common understanding of requirements and affect the requirements elicitation process. For example, there are different system requirements for young people who are familiar with technology and elderly people who need more straightforward software rather than complex and confusing systems [14].

Based on the diverse characteristics of end-users, requirements engineers can extend existing requirements and identify and address acknowledged problems. Therefore, negative and positive emotions, and different cognitive levels and human values should be considered from different perspectives to ascertain their impact on actions [19]. Modelling the diversity of stakeholders, including end-users, and incorporating human factors for specification and management of requirements is a significant challenge in software engineering [10], [41].

Previous work [17] has described how human-centric approaches impact RE during the software development life cycle and has outlined the influence of software development team members inside the project on RE-related activities. Cooperativeness, hardworking competence, transparency, and an interactive and organized approach can have a positive impact on requirements elicitation, prioritization and management. The work in [17] also discusses as one area of future research how involving external end-users can potentially impact the requirements elicitation phase. In addition to the previously described studies, the paper [29] focuses on personality traits and creativity potential and its influence on RE. The study demonstrates how additional requirements for software development can be enumerated through the approach of idea creation process by considering different personalities of Crowd RE including stakeholders.

Recent work [16] presents that the requirements elicitation phase is development phase most affected by human factors. The work highlights the lack of human-centric tools and models based on human factors to improve RE. Some studies [43], [42] demonstrate the lack of focus on the human-centered issues during requirements elicitation, and emphasize the importance of considering such issues in the earlier stages of software development.

In addition to general RE, it is also important to understand the role of the human element and its impact on the development of security systems. Capturing security requirements from a user-centered perspective gives better understanding of the real needs and requirements of end-users, enabling safe and secure software [30], [1].

A study conducted by Alwidian and Jaskolka [3], [42] proposed a framework to consider human-related factors (HRF) in security requirements elicitation by characterizing profiles and cognitive levels of RE team members. The model presents only the profile analysis of requirement engineers and its impact on the requirements elicitation phase while working with security concerns expressed by stakeholders. However, the study highlights as a future work the need for research on profiling of security engineers and stakeholders and how their attitudes and behaviors can impact the quality and validity of the requirements generated.

Other studies [31], [25], have stressed the need to align social concerns with information system security. Research conducted by Naqvi and Seffah [31] presents a requirements engineering methodology that addresses human factors at the requirement and design stage and includes systems designers and experts in the usability and security fields. Their study does not, however, cover the diversity of end-user and attacker perspectives.

In another study Wang and Ni [25] discuss human vulnerabilities and potential attackers, present the social engineering ontologies, and underline the importance addressing the human element in security requirements elicitation.

As demonstrated in extant literature, studies do not, however, extend to presenting a practical methodology for incorporating human factors related to a diverse group of end-users, and attacker profiles into security requirements elicitation.

2.2 The role of personas in RE

Software engineers integrate personas as a design tool into the requirements engineering process for several reasons. Firstly, it helps to categorize a software's stakeholders to better understand their needs, motivation, and behavior from different perspectives of their expectations [39]. This information can enhances the user-centered approach es and assist in developing user-friendly systems and increasing the usability by focusing on user characteristics and capturing various stakeholders [9], [8].

Secondly, use cases are often based on personas and provide detailed information of required functionalities and guide the development process [18]. To organize the development flow, software engineers can extract additional security requirements gathered from use cases and rank the functionality priorities of different stakeholders [21]. Thirdly, personas can be an instrument for interdisciplinary communication involving process stakeholders, usability engineers, business analysts, and security engineers [21],[20].

2.3 The role of personas in security requirements elicitation

By providing information about different types of end-users, their roles and interactions, personas can be useful in terms of identification of possible security concerns and software product vulnerabilities [23], [38]. The utilization of personas in the context of requirements elicitation provides a comprehensive way to capture the motivation of end-users,human behavior, and security concerns, thereby facilitating the identification of risks and vulnerabilities [37]. Since the development process consists of several iteration phases, the complexity of solutions makes it impossible to return to the first step of the development process. The preventive strategy of paying attention to the human aspects by applying a persona-approach provides the opportunities to mitigate the risk of extra steps in the development process.

By implementing a persona-based approach in the security requirements elicitation phase, software engineers can capture different user stories and scenarios which, in turn, can lead to usable and secure systems development [12]. Use case scenarios involving persona assumptions in line with other techniques can improve analysis, and specification, and validation of the technical requirements of a system, giving better understanding of the security context [21]. Additionally, as part of risk analysis, engineers can identify potential threats, vulnerabilities and motivations of malicious users on the basis of attacker persona [5], [28], [12].

Work in [36] presents a persona security technique for enhancing the elicitation of security requirements for safety-critical embedded systems that includes the interconnection between functional and security requirements. Despite the valuable insights in this paper, the study does not explicitly consider user and attacker perspective.

Considered as whole, it is evident that the research gaps remain as regards human-centric implementation in requirements elicitation to enhance and optimize the security requirements elicitation process. The role of human factors in requirements elicitation is significant, for which reason the next section of the paper presents a possible way to implement a human-centric approach in software development that addresses human aspects in security RE and improves the quality of the requirements gathered. By proposing assumptions for persona in the early stage of software, the proposed method responds to some of the challenges of security requirements identification. The inclusion of personas early in the development lifecycle can help to reduce time required for repeated cycles of changing requirements after the initial iteration of the development system, reduce vulnerabilities and counter possible threats.

3 A novel method for persona-driven security requirements elicitation

Motivated by existing challenges in addressing human factors in security requirements elicitation, we created a method of implementing personas across the entire development process that can improve the design of requirements in line with security and usability aspects. A persona-driven approach can also be integrated into the security requirements elicitation phase to help discover unidentified requirements or refine existing functional and non-functional requirements [21], [36]. Based on persona assumptions, the proposed approach identifies different security needs and human vulnerabilities. For instance, a user who frequently uses the system may be reluctant to log out or forget about it, yet, at the same time, there is an expectation of data safety and privacy. This human vulnerability can lead to consider extra security requirements such as user audit activities or set session timeouts.

The proposed persona-driven method is shown in Figure 1.

Software engineers collect requirements from customers before designing and developing software products. This step gives an understanding of the initial requirements of the users. After communication with the customers, the development team creates a software requirement specification (SRS) or product backlog, depending on the project management methodology used, waterfall or agile, to formulate requirements for business needs [11], [34].

In most cases, initial requirements give obvious and general characteristics about the desired solution. Once this information is available, requirement engineers can start to work more precisely with work with security requirements. Here, we introduce the idea of a human-centric approach while working with security requirements.

Software engineers can extract security requirements from the SRS or product backlog and associate potential users or groups of users with these requirements. This data will give an opportunity to develop persona assumptions of software users and enables description of their needs and pain points. The persona development process can be based on providing workshops or interviews. The user persona encompasses characteristics like motivation, culture, age, and demographic group that can effect on software requirements [17], [36]. In addition, the implementation of this approach considers potential attackers and their behavior and motivations based on potential threats posed to the system [5], [28]. Security experts use various frameworks such as STRIDE, OWASP, and PASTA to identify security threats [26]. An understanding of the possible dangers and potential threats to the system being developed supports the creation of an attacker persona who might be capable of or help to find vulnerabilities in the system [5], [28].

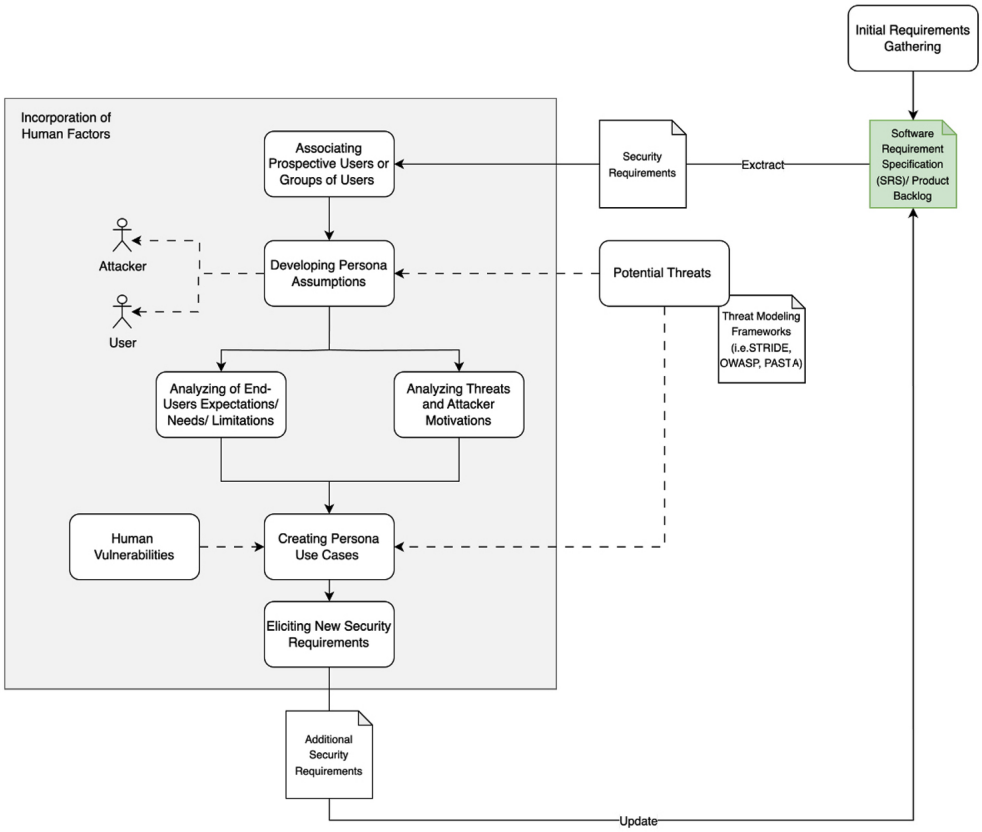

Figure 1: Proposed method for implementation of a persona-driven approach for security requirements elicitation

As a next step, the approach proposes complex analysis of the main goals, needs, limitations and motivations of the user personas and potential threats and motivations of attackers.

Then, considering human factors, vulnerabilities and potential threats to the system, security engineers can create use case scenarios describing the interactions between the system and its actors based on persona analysis from the previous step [13]. This will give scenarios showing a broader perspective of how the system will be used by various users and possible attackers.

Use cases considering different personas assumptions can help explore new security requirements or clarify additions to existing requirements [36]. Thus, the document SRS or product backlog generated will contain more specific requirements, resulting in more secure system development.

This persona-approach allows usability requirements related to security issues and detection of non-functional system requirements to be addressed at the same time, making elicitation of security requirements more human-centric. While the proposed approach is structured way to incorporate human factors and has several benefits for improving security requirements elicitation, it is important to consider that challenges may arise in practical cases. These include potential bias in persona creation, difficulties scaling across large or distributed development teams, and integrating persona-based ideas into existing security engineering workflows. The acknowledgement of potential limitations provides a foundation for the further research and development of strategies in practice for more widespread adoption.

4 Discussion on the application of persona-driven security requirements across the software development lifecycle

Security requirements elicitation at an early stage impacts the entire development lifecycle [3]. In the requirements elicitation phase, personas can be created to describe user needs and potential risks and vulnerabilities [20]. Based on the attributes of the personas, the development team can identify key personas, their expectations of the product, motivations for use, and assumptions of product security.

During the software design and the development phase, requirement engineers can apply the security expectations and requirements gathered from the identified personas at the initial stage in security measures for protection against the vulnerabilities [36]. This knowledge also underlines the design of use-case models, for instance, how different users will interact with the system and their behavior from a security perspective [13]. Furthermore, by modeling the activities and behavior of the attacker personas, the development team can integrate hacking or malicious activity scenarios

Table 1: Example of security requirement elicitation via the persona-driven approach

Category	Description
Persona	**Peter** **Age**: 35. **Job Description**: Account manager in marketing agency. **Personality**: Curious, adaptable, sometimes impatient. **Behavior**: He works remotely 2 days per week from different devices. During working hours, he receives corporate emails and a lot of external clients' information. Sometimes he uses public Wi-Fi while at an airport or on a train. **Security Attitude**: He is not tech-savvy but follows company's security standards if they are not overly complicated. **Goal**: Be available to managing external
Potential (Human) Vulnerabilities	PV1: May use weak or reused passwords due to using different programs and password fatigue. PV2: Often works in public places. PV3: Can forget to update software. PV4: May ignore browser security warnings to complete tasks quickly.
Potential Threats	PT1: Attacker intercepting session while using public Wi-Fi. PT2: Phishing attacks via email. PT3: Account hacking due to weak password protection. PT4: Date leakage when exchanging data with external customers.
Security Concerns	SC1:The system should prevent accidental data leakages for non-recipients; SC2: Work from different devices and unsecured network should be protected.
Usability Concerns	UC1: The process of sharing files should be less bureaucratic and should not involve a lot of steps. UC2: The employee should not be overstressed due to over complicated safety policies and rules.
Additional Security Requirements	**Authentication:** The system should have MFA as an additional verification if login is from an unrecognized device or public network. **VPN Integration:** The system must use incorporation of a secure tunnel when logging in from public Wi-Fi. **Phishing Awareness Notifications:** Phishing Awareness Notifications: The system should provide security tips when suspicious activity is detected. **Session Log Out:** The system must have an option for users to immediately determinate all active sessions from all devices. **Password Manager Integration:** Password Manager Integration: The software must support password managers. **Session Timeout:** The system must have a timeout after 15 minutes per user session. **Access-Restricted Sharing:** The user should be able to share sensitive data with external clients by sharing files with different restrictions (i.e. limited by time, role, number of available views).

[27]. Thus, the improvement in security requirements elicitation can help counter threats by developing stronger preventive security strategies.

The implementation of personas and their persona-based behavior in a variety of situations can be used in the testing and deployment phases to analyze security scenarios [5]. For example, they can help test vulnerabilities in developed software while testing social engineering cases on persona assumptions. The maintenance phase aims to keep the software updated to meet customers' needs by modifying and updating the software features. It gives an opportunity to assess the impact of personas' vulnerabilities, prioritizes them and creates new security strategies for the next cycle.

After each stage of the software development process, following the proposed approach, the authors suggest applying practice for updates and improvements based on personas. Information about personas and identified activities or risks help requirement engineers to maintain a relevant and actual list of security requirements.

We present an example scenario of implementation of a persona-based approach implementation for security requirements engineering in the software development process. The initial security requirements are described in Figure 2 and possible elicitation of additional requirements via end-user persona is presented 1.

SR1: The system must support two-factor authentication (2FA) for all users.
SR2: All sensitive data must be protected from unauthorized access.
SR3: User accounts should be locked after five failed login attempts.
SR4: Sessions must be terminated after 15 minutes of inactivity.

Figure 2: Initial security requirements

Human factors. Personas represents the motivations and expectations of a group or individuals. They will help to understand what drives users and their goals for using software. In line with user characteristics, personas give a description of the level of tech literacy and skills [21]. This has a direct impact on how users interact with software products, thereby facilitating the overcoming of challenges associated with a misunderstanding of security mechanisms. Moreover, software engineers can develop use cases for different scenarios which will cover a user's behavior and his interactions with the final product in curtain real conditions [4]. Usability aspects. The persona-approach will help create intuitively understandable interfaces with easy-to-use functionalities for their end-users. Considering the diversity of users, their experiences, and their limitations, the analysis of use cases help engineers follow end-users' paths, understand their needs and even detect potential weak areas in the software to create more usable solutions [13]. In addition, continuous feedback based on personas from the earliest stages can improve the quality of features and design.

Security aspects. By implementing a persona-based approach, requirement engineers can identify security requirements, detect potential vulnerabilities, and model behaviors and threat scenarios at an earlier stage in the development process [32]. This makes it possible for the development team to initially plan preventive measures and detect possible software vulnerabilities by modeling hacking scenarios or actions of malicious users [5]. Additionally, the development of attacker personas will help prevent the need for software refinement and full revision of the system at late stages caused by system problem areas identified during the testing and deployment stages [28]. Persona-based testing and adjustments enable elicitation of additional security requirements during the development process which leads to a reduction in late-stage refinements costs.

The findings of this research provide the insights of human-centric approach to enhance security requirement elicitation. The paper presents novel idea based on the background and related work, however, it is limited by validation and evidence of practical use. As a future work we propose evaluation of persona impact in security requirement elicitation, validation of proposed method in the practical cases, creation and testing attacker scenarios models for developing preventive strategies.

5 Conclusion

This paper presents a persona-driven method for software development cycle to enhance the security requirements elicitation in early stages of the development cycle. The approach consists of 1) identification of potential users/ attackers, 2) development of persona assumptions, 3) description of possible scenarios in use cases, 4) elicitation of refinements or additional requirements, and 5) a continuous refinement process for keeping requirements up-to-date.

The presented method gives the security engineers the opportunity to engage stakeholders in security requirements elicitation, allows identification of potential vulnerabilities of systems, and enables a reduction in repeatable cycles of product refinement by characterizing groups of users or individuals based on their behavior.

Compared to existing methods and frameworks, the paper presents a novel approach for integrating personas into security requirements elicitation. By incorporating human factors into the early stages of software development, this persona-based approach helps to fill a gap in literature and opens up opportunities for further research.

In future work, studies should be extended to include more comprehensive research to incorporate received findings from the literature review with practical validation. It is intended to conduct workshops or primarily interviews with practitioners to validate from the industry side the approach proposed in the paper and gain more detailed understanding of persona assumptions. Moreover, future work could focus on validation of the persona-approach by integrating its use into practical cases.

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